Intertemporal Interaction and Symmetric Difference Learning for Remote Sensing Image Change Captioning

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***Abstract*— Remote sensing image change captioning (RSICC) is more challenging than remote sensing change detection task, which requires extracting occurred changes in similar remote sensing image (RSI) pairs while generating change caption. However, few works have been investigated on RSICC, the main challenges come from how to learn abundant change clues and face the modality gap. To handle these problems, we rethink this task from the perspective of obtaining and aligning symmetrical change features for temporal RSIs. In this work, the proposed intertemporal interaction and symmetric difference learning network are cascaded through several multitemporal integration units to model differences from coarse to fine representations. Specifically, we design a cross-temporal attention (CTA) mechanism to probe direct interaction between bi-temporal RSIs for motivating information coupling between intralevel representations and suppressing irrelevant interferences. To learn robust change features, a symmetric difference transformer (SDT) module is devised to guarantee temporal symmetry between the "before-to-after" and "after-to-before" change representations. Besides, the bi-directional triplet ranking loss is adopted to guide the network to learn strongly discriminative and temporal-symmetric change representation. Extensive experimental results on Dubai-CC and LEVIR-CC datasets demonstrate that our framework with the proposed components can achieve excellent performance and surpass recent state-of-the-art methods. https://github.com/romanticLYP/TISDNet**

***Index Terms*— Cross-temporal attention (CTA), remote sensing change captioning, symmetric change detection, temporal remote sensing images (RSIs).**

I. Introduction

THE same Earth's surface is continually changing over multiple periods. Benefiting from the development of remote sensing technology and multitemporal remotely sensed Earth observation, multitemporal and high-resolution remote sensing images (RSIs) have become available. Automatically identifying changes in the same region in RSIs has developed rapidly from data accumulation. The existing change detection approaches can be generally divided into binary change detection and semantic change detection. Specifically, change information obtained by binary change detection methods will be visualized with binary values, semantic change detection networks not only locate changed fields but also identify change types. For example, the spatiotemporal enhancement and interlevel fusion network (SEIFNet) [1] is introduced for remote sensing change detection tasks, which improves the ability of target-level change feature representation. These change detection works only recognize the change regions or output pixel-level change maps with separate classes, but ignore the relationships and semantic differences of changed objects in a pair of "before" and "after" RSIs. It is still difficult for users to directly access and understand important change information. In particular, RSI change captioning (RSICC) aims to generate comprehensive sentences that comprehend the contents of the RSI pairs and summarize their difference in language level [2], [3], [4], as shown in Fig. 1. The change captioning techniques for RSI play an important role in urban planning [5], [6], environmental monitoring [7], [8], [9], and disaster monitoring [10], [11], [12].



Ground Truths:
1. Many buildings appear alongside the road.
2. Villas are built on the bareland along the road.
3. Many villas and vegetation have appeared on the bareland.
4. Some houses are built at the corner of the road.
5. Many buildings appear around the road.

Fig. 1. Example from LEVIR-CC dataset of RSICC about new building change. The five annotations on the right side are the corresponding differences between a pair of RSIs. The proposed RSICC approach aims to observe and describe the change with natural language.

In the past few years, few methods have been designed for natural image change captioning (NICC) [13], [14], [15]. Generally, the exiting NICC models widely adopt encoder–decoder framework. Convolutional neural networks (CNNs) encode RSI pairs, and the ability to generate sentences is achieved by long short-term memory (LSTM) or transformer decoder. In addition, spatial attention variants and self-attention applied

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in NICC models have been proven their salience. The significant inspiration can promote the development of RSICC tasks. For example, to highlight differences in dual visual representations, a simple dual dynamic attention (DUDA) was proposed by Park et al. [13], and the change map was obtained through the subtraction strategy of the bi-temporal features. However, the relationships of vision and semantic between objects are not sufficiently concerned. To overcome these limitations, Tu et al. [15] proposed the self-semantic and cross-semantic relation block to detect the changed areas, the change captioning model gets profit from its advantage of building more relationships within intermodality and intramodality. Furthermore, Qiu et al. [14] applied two transformer-based encoders to learn the correlation between different image regions for detecting multiple changed parts, while the transformer decoder can better model the correlation between change regions and sentences than the LSTM structure. Similarly, Yue et al. [16] constructed the geometry-semantic interaction refining to tap the potential of semantic and positional interactions of intraimage, then the reliable difference representation could be obtained by designed hierarchical interaction.

Although the aforementioned methods have achieved success in the NICC task, the specialized nature of the RSICC task makes change information trouble. Especially, from the "View of God," the challenges of RSI are scale ambiguity, category ambiguity, and rotation ambiguity [17], [18], as well as the semantics in RSI become more complex than natural image. Directly integrating the NICC models with RSICC work will not bring improvement in caption performance, because change features are weakly related to the scene of input RSIs. Therefore, the problem of exploiting a suitable structure for the RSICC task is of great significance and remains an urgent development.

Recently, some researchers have studied the RSICC and generated sentences to distinguish relevant scene changes. Pioneer's work [19] combined a pretrained CNN with classic recurrent neural networks or support vector machines to describe the possible changes, the differences are captioned with the distraction of bi-temporal visual features or image-level change. Later, a large-scale dataset named LEVIR-CC was released to solve the issues of small-scale datasets of LEVIR CCD and Dubai CCD [19]. Particularly, a dual-branch transformer encoder [20] captured and recognized multiple changes of interest, and then a transformer caption decoder generated sentences describing semantic differences. This indicates that the transformer-based architecture is also effective in RSICC tasks. Similarly, Chang and Ghamisi [21] proposed dual hierarchical self-attention blocks to judge if the change has actually happened in RSI pairs. Inspired by the utilization of multiscale information in other remote sensing tasks [22], [23], Liu et al. [24] considered embedding multiscale visual features of the "before" and "after" RSI pairs, which strengthened the cognition of semantic change to perform RSICC tasks better. In addition, Zhou et al. [25] designed a pretrained single-stream feature extractor and difference extractor which extracted the difference features explicitly. In order to obtain the most critical changes of interest guidance, Cai et al. [26] integrated an interactive change-aware encoder into a novel change caption decoder which selected essential discriminative different features to improve the change captioning performance.

Despite progress, there exist some deficiencies that desire to be explored.

1) Due to the uncertain change distribution in bi-temporal RSIs, a better change-relevant representation is helpful for exploring deeper semantic changes about changed objects in two RSIs. Although self-attention and transformer in encoder can effectively learn visual relationships in intraimage, the task-irrelevant interference will be reserved and limit the difference acquisition.
2) For change representation, most researchers squander excessive energies on learned bi-temporal features, then use subtractive operation or similarity calculation to judge whether there is a change. Strongly discriminative change representation is insufficient exploitation in these works, which is a critical and exclusive point of the RSICC task. Therefore, more emphasis should be put on the investigation of valuable change features.

In this article, we propose an intertemporal interaction and symmetric difference learning network to tackle the above two problems. Specifically, we first design a novel cross-temporal attention (CTA) mechanism to tailor interaction between CNN-based bi-temporal features. Similarly, the multilevel spatiotemporal difference enhancement modules in SEIFNet [1] were designed to detect local contextual information for supplementation and reduce the noise of interference, which consists of a subtraction branch and a connection branch. However, in our attention strategy, the coupled bi-temporal features with concatenation are adopted as the query item. Thus, CTA can model the semantics and relationships interaction for changed regions and suppress the task-irrelevant information. Furthermore, a symmetric difference transformer (SDT) module aims to accurately learn difference representation, in which the semantic change is derived from the "before-to-after" and "after-to-before" change features. We gain the final difference representation by concatenating the symmetric difference features, while an adaptive context fusion module in SEIFNet [1] was applied for coarse-to-fine change regions between bi-temporal feature maps. Note that the bidirectional triplet ranking loss serves as an effective supervision to guarantee temporal symmetry on bi-direction change features. Finally, the transformer decoder generates a sentence that can describe the happened changes.

A. Contributions

To sum up, we have made the following four contributions in this article.

1) A novel intertemporal interaction and symmetric difference learning network are proposed for the RSICC task. Compared with the traditional change detection process, our model introduces high-quality symmetric change representations, which can preserve change details between bi-temporal RSIs, rather than relying solely on shadowy and abstract change information.

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2) Following the CNN-based features extracted from the input RSI pair, an intertemporal CTA module is proposed to aggregate the genuine changes. It harmonizes the advantages of self-attention and cross-attention to derive information interaction and leads to enhanced representations of bi-temporal RSIs with strong separability and suppressed noise.
3) To leverage temporal symmetry in the change features detection process, the SDT block is devised in cascaded multitemporal integration units, in which the change consistency between the "before-to-after" and "after-to-before" change features is considered. It can significantly improve the ability to change representation.
4) For constraining the consistency of bi-directional change features, an alignment supervision strategy enforces change part correspondence while depressing unchanged regions. Specifically, the bi-directional triplet ranking loss is adopted to measure the similarity between the temporal-symmetric change representations.

B. Organization

The rest of this article is organized as follows. Some previous related studies on RSICC tasks are introduced in Section II. In Section III, the proposed framework with designed attention modules and how to learn robust change features are introduced in detail. Section IV presents the performance of the proposed method and discusses experimental results on two public datasets. Finally, our conclusions and future works are given in Section V.

II. Related Work

The task of RSICC has many similarities with remote sensing image captioning (RSIC), while the former focuses on changes between RSI pairs, the latter generates descriptions for a single RSI. In this section, we briefly introduce and discuss some recent studies on RSICC.

Before introducing the works of RSICC, we first review some representative works in the RSIC task. The RSIC aims to generate a sentence to describe a single RSI, which has been intensively studied for several years. It is important for the RSIC task that how to enhance the capability of complex cross-modal reasoning through visual and semantic multilevel interaction. Recently, transformer architecture has achieved excellent performance on RSIC task [27], [28]. In general, "CNN-transformer" [29] and "transformer–transformer" [30], [31], [32], [33] RSIC frameworks are strong approaches compared with "CNN-LSTM" algorithm. This is because the transformer has the significant advantage of using self-attention to learn long-range dependencies at the level of intraimage or intratext. For example, the model designed by Zia et al. [34] is related to method [29]. For employing semantic words in a transformer-based encoder, the mask-guided transformer network [31] took the raw RSI as input for generating the cross-modal representations, including semantic information from topic token and patch-level visual features, which shows great potential in a transformer-based decoder. The aim of these methods is to form a more powerful feature representation and utilize attention mechanisms (i.e., spatial attention and self-attention) to perform vision-to-linguistic transformations.

RSICC task is a new branch of the vision-language understanding task in the remote sensing field. Although many methods have been proposed for RSIC, only a few studies have focused on RSICC which requires locating change features between well-aligned RSI pairs and describing their difference with a sentence. This is because there is not an accredited dataset like RSICD for the RSIC task. Hoxha et al. [19] adopted two change captioning systems to generate sentences for occurred changes. In this method, image-level changes and feature-level changes are utilized to present possible changes in bi-temporal RSIs, and the change features will then guide the decoder to predict the word step by step. This is the earliest work for the RSICC task. In addition, two small RSICC datasets, namely, LEVIR CCD and Dubai CCD, are published for further research. Furthermore, Liu et al. [20] applied a transformer structure to constitute the RSICC network: in the encoder, a dual-branch transformer helps to better model change features on RSI pairs; in the decoder, the transformer can filter out irrelative attention results for generating change description. The observable improvements prove that the transformer structure is adaptive in the RSICC task. In addition, they create and publish a big new RSICC dataset, LEVIR-CC, to augment LEVIR CCD. This new dataset contains more RSI pairs and more sentence patterns than the Dubai-CC dataset. Noteworthily, some strategies have also been used to improve performance, such as incorporating multiscale information and stacked hierarchical self-attention. To alleviate the scale-diversity problem in input RSI pair, a pure transformer-based RSICC model with multiscale bi-temporal visual features was proposed by Liu et al. [24]. The multiple progressive difference perception layers combined with a scale-aware reinforcement module were stacked with a transformer-based encoder–decoder, which progressively exploits the differencing features based on bi-temporal multiscale features. However, when the captured difference feature is not what the model really expects, the attention module considered by Chang and Ghamisi [21] could give satisfied results. Thus, a novel changes-to-captions network was proposed (Chg2Cap) for accurately describing changes occurring in multitemporal RSIs, which comprised three components: a Siamese CNN feature extractor, an attentive encoder, and a transformer-based caption generator. Note that the model applying a hierarchical self-attention block in the encoder can dynamically locate change-related features. Thus, the effects brought by change features can be further improved. Zhou et al. [25] developed a novel single-stream extractor network (SEN) with contrastive pretraining which faced high computational costs under the dual-stream extractor. Then, it forms an explicit difference feature encoder to better model the difference information for the RSICC task. Later, a lightweight transformer was proposed by Sun et al. [35], which contained a high-level features extractor,

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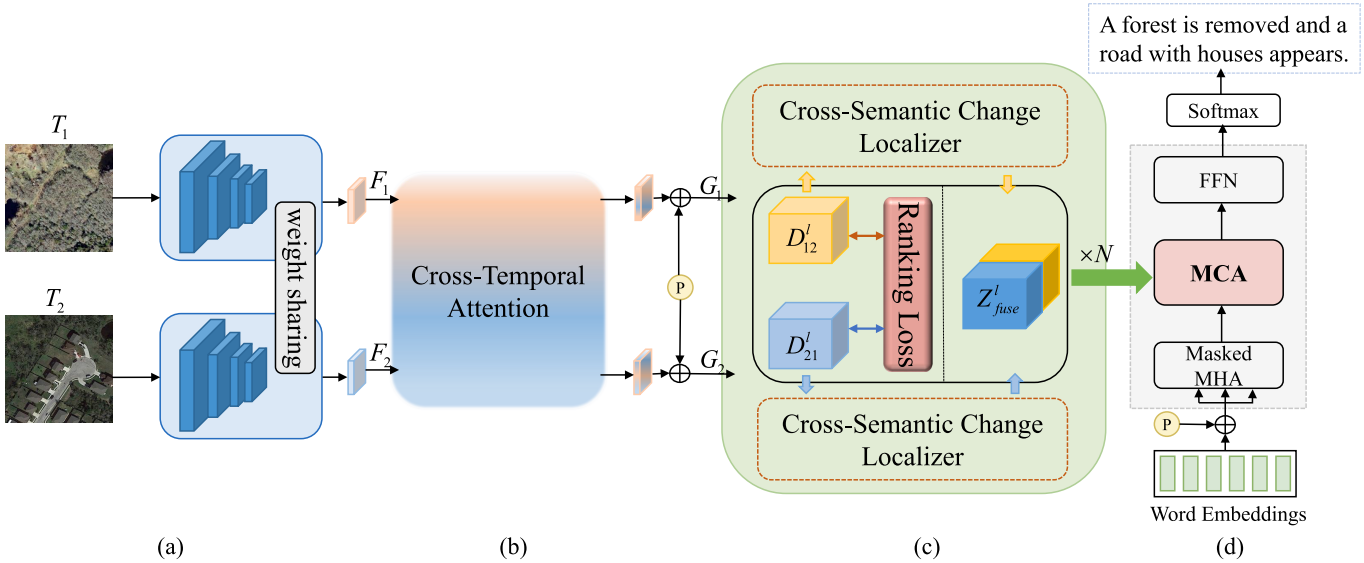


Fig. 2. Overview of the proposed intertemporal interaction and symmetric difference learning for RSICC task. Given a pair of RSIs, the Siamese feature extractor first extracts original temporal features. Next, the CTA module is applied to enhance the spatial and semantic relationships of the bi-temporal features. After that, the enhanced temporal features are fed into the SDT module for centering more on the genuinely changed information. Finally, the transformer decoder describes the changes. (a) Feature extractor. (b) CTA. (c) SDT. (d) Caption generator.

sparse focus attention, and description decoder. The efficient multiple spatial-temporal difference modeling can guide the decoder to accurately focus on changed contents during change captioning [26], [36]. For example, Cai et al. [26] proposed an interactive change-aware transformer network (ICT-Net) that an interactive change-aware encoder was employed to extract relevant multiscale changes, where they used multilayer adaptive fusion module prompting designed transformer for change caption generation.

However, the aforementioned methods capture changed features between two RSIs mainly based on direct subtraction and global similarity, while not trying to learn the interaction between bi-temporal features before accessing changes. In addition, the problem of the temporal symmetry in change detection is disregarded. Instead, we propose an intertemporal interaction and symmetric difference learning network to accurately locate changes between bi-temporal features and describe the occurred changes with a transformer generator. It fulfills two RSI interactions at the feature level and then learns symmetric change information. Moreover, we employ the bi-directional triplet ranking loss to constrain the change consistency.

III. Methodology

Here, we focus on attractive and efficient change learning for solving the coarse change representations in the RSICC task. Our motivation mainly comes from that there exists semantic change-dependent information among objects and the temporal-symmetric relationship in bi-temporal RSIs. Therefore, our aim is to resist irrelevant changes for progressively inferring change location. As shown in Fig. 2, the architecture of our method consists of three parts: 1) a CTA mechanism distills task-relevant information; 2) an SDT learns symmetric difference features, optimizing by a bi-directional triplet ranking loss; and 3) a transformer decoder translates genuinely changed features to sentence.

A. CTA Mechanism

Formally, given two aligned RSIs of "before" I_{bef} and "after" I_{aft}, which are collected from time T_1 and time T_2, respectively. We use Siamese CNN [37] to construct a temporal feature extractor. Suppose that the features of two input RSIs are represented as (F_1, F_2), which is defined as $F_1 = f_1(I_{\text{bef}})$ and $F_2 = f_2(I_{\text{aft}})$. Note that $f_1(\cdot)$ and $f_2(\cdot)$ represent the ResNet101 feature extractor with weight-sharing, (F_1, F_2) features usually contain the context information around objects. And if an object changed in "after" I_{aft}, it means that semantic changes have happened.

In general, two parallel visual features [19], [20], [21] were directly used for the difference feature map. It is deficient in learning the relationship of semantic, spatial, and location in the process of change detection. The well-known self-attention mechanism [38] can boost feature learning by focusing on important parts of original features in other vision tasks. However, it is not applicable for analyzing change-related information in pair features independently. In this work, the obtained pair features (F_1, F_2) mainly contain texture and detailed information of ground objects, which also integrate some unrelated information. To further take the semantic relationships around objects into account, we design the CTA mechanism following the Siamese CNN structure, which can filter invalid representations and fulfill feature interactions between bi-temporal features. As illustrated in Fig. 3, the pair features (F_1, F_2) are transformed by 1×1 convolution for three independent features, i.e., the query Q, key K and value V. They can be computed as

$$Q_i, K_i, V_i = f_{qi}(F_i), f_{ki}(F_i), f_{vi}(F_i) \tag{1}$$

where $f_{qi}(\cdot)$, $f_{ki}(\cdot)$, and $f_{vi}(\cdot)$ denote independent 1×1 convolution, and $i = 1, 2$. To capture the complex and semantic relations about changed objects, a simple method interacts with bi-temporal feature (F_1, F_2) for most change-level

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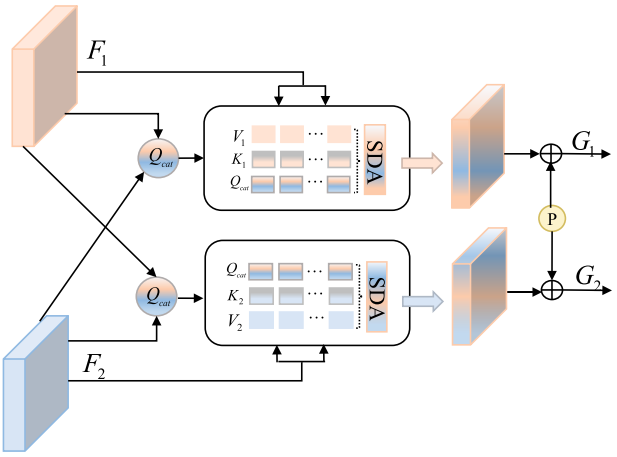


Fig. 3. Illustration of CTA module, in which the scaled dot-product attention (SDA) models the semantic relation in bi-temporal features.

relevant information, preserving the high-level information around changed areas as much as possible. Since Q_1 and Q_2 also contain features details of bi-temporal RSIs, which are concatenated into $Q_{\text{cat}} = [Q_1, Q_2]$ to construct the complementary queries. Then, CTA on both V_1 and V_2 can then be simultaneously derived by SDA for modeling the semantic relation or passing message of bi-temporal features. The dot-product operation is used to calculate the similarity between Q_{cat} and K_1 or K_2 by

$$\alpha_i = \text{softmax}\left(\frac{Q_{\text{cat}} K_i^T}{\sqrt{d_i}}\right) \tag{2}$$

where d_i is a scaling factor, and $i = 1, 2$. We perform weighted summation to obtain the attended value $\alpha_i V_i$, which can glimpse the representation of another RSI input. The output of the CTA layer is defined as follows:

$$C_i = \alpha_i V_i + V_i \tag{3}$$

where $i = 1, 2$. Note that C_i incorporates semantic relationships and difference representations related to bi-temporal RSIs, which efficiently bridges the information interaction and exploits the temporal contextual characteristics. To improve the perception ability of position change in the RSICC task, the spatial information of each RSI can be augmented by injecting a position embedding into the temporal features C_i. Therefore, learnable position embedding assigns a constant token to each feature as follows:

$$G_i = C_i + F_{\text{pos}} \tag{4}$$

where $i = 1, 2$, and F_{pos} denotes learnable position embedding.

B. Symmetric Difference Transformer

Simple subtraction is generally adopted between "after" and "before" features for change representation. However, the "after-to-before" change features cannot sufficiently explore the interfeature difference and incorporate each other's strengths, which will weaken the change representation ability in temporal difference. To overcome the uni-directional change detection problem in some recent RSICC research [19], [20], [21], we propose SDT, which models temporal symmetry for change passing based on multilevel temporal features. The details of the temporal symmetric transformer module are shown in Fig. 4. SDT consists of N identical blocks, a symmetric difference learning, a cross-semantic change localizer, and a multilevel aggregation layer, these are the main components in each block.

1) Symmetric Difference Learning: Given the input temporal features Z_1^{l-1} and Z_2^{l-1} into the lth $(1 \sim N)$ SDT block, a symmetric difference learning given in Fig. 4(a) is constructed for bi-directional change features of the "before-to-after" and "after-to-before." Note that $Z_1^0 = G_1$ and $Z_2^0 = G_2$. For "before-to-after" change features, we calculate the corresponding measure on each location between the feature pair

$$\Psi\left(Z_1^{l-1}, Z_2^{l-1}\right) = \text{sig}\left(W_{\text{bfa}}^l Z_1^{l-1} + W_{\text{bfb}}^l Z_2^{l-1}\right) \tag{5}$$

where W_{bfa}^l and W_{bfb}^l are learnable parameters, sig$(\cdot)$ is activation function Sigmoid. There will be a stronger response if no change appears in the same location. Based on the active signal $\Psi(Z_1^{l-1}, Z_2^{l-1})$, we can learn the unchanged information in Z_2^{l-1} as follows:

$$\tilde{Z}_2^l = \Psi\left(Z_1^{l-1}, Z_2^{l-1}\right) \odot Z_2^{l-1} \tag{6}$$

where $\odot$ is element-wise multiplication. Finally, the difference for "before-to-after" is captured by subtracting $\tilde{Z}_2^l$ from Z_1^{l-1}

$$D_{12}^l = Z_1^{l-1} - \tilde{Z}_2^l. \tag{7}$$

Since the predicted unchanged information $\tilde{Z}_2^l$ and changed features D_{12}^l are only with respect to one kind difference of "before-to-after," it cannot exploit the causal relationship between semantic representation and change representation which is not in the "before-to-after." For an efficient model, the convincing change representation between bi-temporal RSIs should be temporal-symmetric. Thus, it should also capture the difference of "after-to-before," which is captured as follows:

$$\Psi\left(Z_2^{l-1}, Z_1^{l-1}\right) = \text{sig}\left(W_{\text{afa}}^l Z_2^{l-1} + W_{\text{afb}}^l Z_1^{l-1}\right) \tag{8}$$

where W_{afa}^l and W_{afb}^l are learnable parameters. With $\Psi(Z_2^{l-1}, Z_1^{l-1})$, we can learn the unchanged information in Z_1^{l-1} as follows:

$$\tilde{Z}_1^l = \Psi\left(Z_2^{l-1}, Z_1^{l-1}\right) \odot Z_1^{l-1} \tag{9}$$

where $\odot$ is the element-wise multiplication. Therefore, the difference for "after-to-before" is captured by subtracting $\tilde{Z}_1^l$ from Z_2^{l-1}

$$D_{21}^l = Z_2^{l-1} - \tilde{Z}_1^l. \tag{10}$$

Apparently, the symmetric change features (D_{12}^l, D_{21}^l) contain the similar semantic-change causal relationship. To guarantee $D_{12}^l = D_{21}^l$, symmetric supervision can be implemented to improve the robustness of symmetric change detection. To reduce computational complexity, applying an adaptive-avg layer on (D_{12}^l, D_{21}^l) is designed to extract symmetric global change feature (d_{12}^l, d_{21}^l). Subsequently, the bi-directional triplet ranking loss [39], [40] is adopted to supervise the

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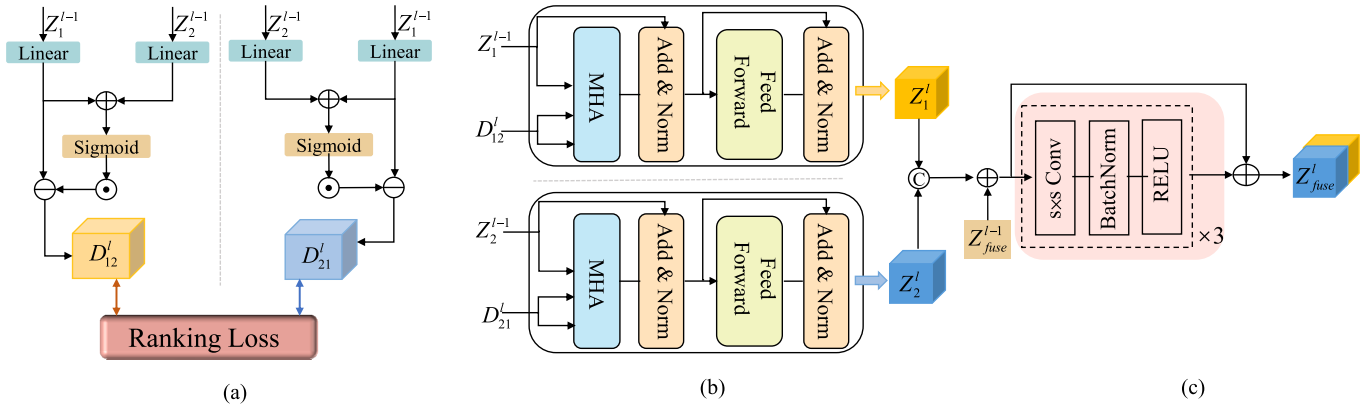


Fig. 4. Detailed architecture of the on layer in SDT block, which consists of (a) symmetric difference learning, (b) cross-semantic change localizer and (c) multilevel aggregation.

training of the symmetric change process for aligning the symmetric vectors (d_{12}^l, d_{21}^l)

$$L_r^l(d_{12}^l, d_{21}^l) = \sum \left\{ \begin{array}{l} \max\left[0, \gamma - s\left(d_{12}^l, d_{21}^l\right) + s\left(d_{12}^l, d_{21}^{l-}\right)\right] \\ + \max\left[0, \gamma - s\left(d_{21}^l, d_{12}^l\right) + s\left(d_{21}^l, d_{12}^{l-}\right)\right] \end{array} \right\} \quad (11)$$

where γ is a predefined margin parameter, and $s(\cdot)$ denotes the cosine distance function. Given the pair (d_{12}^l, d_{21}^l) and (d_{21}^l, d_{12}^l), its corresponding negative pairs are denoted as (d_{12}^l, d_{21}^{l-}) and (d_{21}^l, d_{12}^{l-}), respectively. The final bi-directional triplet ranking loss of the SDT module is defined as follows:

$$L_r = \frac{1}{N} \sum_{1}^{N} L_r^l(d_{12}^l, d_{21}^l) \quad (12)$$

where N is the count of stacked layers in the SDT block. In our experiment, N is set as 3.

2) Cross-Semantic Change Localizer: The main structure of the cross-semantic change localizer is shown in Fig. 4(b). Like the transformer block, the cross-semantic change localizer is built upon multihead attention (MHA) and feed-forward network (FFN), in which two components have residual connections among them. After generating the bi-directional difference representation (D_{12}^l, D_{21}^l), the cross-semantic change localizer localizes the changed feature in Z_1^{l-1} and Z_2^{l-1}, respectively. For the top branch, Z_1^{l-1} and D_{12}^l are as input. Specifically, D_{12}^l is used as the key and value items, while Z_1^{l-1} is adopted as the query item for the attention strategy

$$Z_{\mathrm{TM}}^l = \mathrm{MHA}_{\mathrm{top}}^l\left(Z_1^{l-1}, D_{12}^l, D_{12}^l\right) \quad (13)$$

while the FFN layer is utilized to process the output of the MHA layer additionally. The output of the top cross-semantic change localizer is defined as

$$Z_{\mathrm{TA}}^l = \mathrm{AN}\left(\mathrm{FFN}\left(\mathrm{AN}\left(Z_1^{l-1} + D\left(Z_{\mathrm{TM}}^l\right)\right)\right)\right) \quad (14)$$

where $\mathrm{AN}(\cdot)$ denotes the composition of residual connection and normalization and $D(\cdot)$ indicates embedded by Dropout function. In the process of locating the difference in the down branch, both Z_2^{l-1} and D_{12}^l should be fed to determine whether the change has occurred in Z_2^{l-1}. The operation of MHA over given inputs is

$$Z_{\mathrm{DM}}^l = \mathrm{MHA}_{\mathrm{down}}^l\left(Z_2^{l-1}, D_{21}^l, D_{21}^l\right). \quad (15)$$

An output of the down cross-semantic change localizer is given by

$$Z_{\mathrm{DA}}^l = \mathrm{AN}\left(\mathrm{FFN}\left(\mathrm{AN}\left(Z_2^{l-1} + D\left(Z_{\mathrm{DM}}^l\right)\right)\right)\right). \quad (16)$$

Finally, we obtain a fusion difference representation via concatenation operation on Z_{TA}^l and Z_{DA}^l

$$Z_A^l = \left[Z_{\mathrm{TA}}^l; Z_{\mathrm{DA}}^l\right] \quad (17)$$

where [;] denotes concatenation.

3) Multilevel Aggregation: The purpose of the aggregation operation is clear; the multistage symmetric change clues are more meaningful than single-stage change maps, which can be carried on throughout the difference learning process. Moreover, skip connections between Z_{fuse}^{l-1} and Z_{fuse}^l are introduced. So that refined change features can be excavated, the flowchart is given in Fig. 4(c). In detail, Z_A^l from cross-semantic change localizer and Z_{fuse}^{l-1} from $l-1$ layer in SDT module are adopted to improve the detail preservation ability of the multilevel aggregation. This process can be formulated as follows:

$$T_{\mathrm{add}}^l = Z_A^l + Z_{\mathrm{fuse}}^{l-1}. \quad (18)$$

As shown in Fig. 4(c), it contains three subblocks and each block performs feature fusion with a convolutional layer, BatchNorm (BN), and activation (i.e., RELU). Mathematically, the overall multilevel aggregation can be formulated as follows:

$$Z_{\mathrm{fuse}}^l = \mathrm{MA}_3\left(\mathrm{MA}_2\left(\mathrm{MA}_1\left(T_{\mathrm{add}}^l\right)\right)\right) + T_{\mathrm{add}}^l \quad (19)$$

$$\mathrm{MA}_s = \mathrm{RELU}(\mathrm{BN}(\mathrm{Conv}_s(X))) \quad (20)$$

where X is the input of the subblock in multilevel aggregation, $\mathrm{Conv}_s(\cdot)$ is the 2-D convolutions with $s \times s$ kernels, and $s = 1, 2, 3$. Such change feature Z_{fuse}^l allows our model to be focused on the genuinely changed regions and has a serious impact on the transformer-based decoder.

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C. Transformer Decoder

As for the decoder, the modality of the change feature and textual feature is different. To enhance the encoding-decoding process, the word embedding features and generated change features are embedded into a single transformer-based decoder to generate a change caption. The decoder is composed of a masked MHA, multihead cross attention (MCA), and an FFN sequentially. Specifically, the MCA performs the change features and textual features to make interaction between them. Based on the final change features $Z_{\text{fuse}} = Z_{\text{fuse}}^{l=3}$ extracted by SDT, the decoder process can be formulated as follows:

$$y_t = \text{softmax}(\text{Transformer}(\omega, Z_{\text{fuse}})) \tag{21}$$

where ω is the embedding of known sentence fragments, the softmax probability is finally mapped to the word dictionary.

D. Loss Function

To train our model, we design loss function L as follows:

$$L = L_{\text{CE}} + \lambda_1 * L_r \tag{22}$$

where L_{CE} is to supervise caption generation, L_r is a bi-directional triplet ranking loss to provide auxiliary assistance for capturing more effective change features, and the hyper-parameters λ_1 is set with 0.2.

In general, the caption generation is trained using the cross-entropy objective function in most RSIC [3], [18], [41] and RSICC methods [13], [19], [24]. Given a target ground truth caption sequence $y_{1:T}^*$ and a captioning model with parameters θ, L_{CE} tries to minimize the sum of the negative log-likelihood of the correctly predicted words at each step given as follows:

$$L_{\text{CE}} = \sum_{t=1}^{T} -\log\left(p_\theta\left(y_t^* \middle| y_{1:t-1}^*\right)\right) \tag{23}$$

where y_t^* denotes a one-hot vector for the nth word in a ground truth sentence of length T.

IV. Experiments

To demonstrate the effectiveness of our proposal, we choose two public RSICC datasets for testing. First, we introduce the details of two datasets, namely, the Dubai-CC dataset [19] and the LEVIR-CC dataset [20]. Then, the implementation details are described. Subsequently, the experimental results of comparisons with state-of-the-arts on two datasets are presented, we also perform ablation experiments to investigate the importance and contributions of the designed modules. Finally, the influence of each factor on the proposed method is analyzed and summarized.

A. Experimental Setup

1) Datasets: **Dubai-CC Dataset** is extracted from the Enhanced Thematic Mapper Plus (ETM+) sensor onboard Landsat 7, on May 19, 2000, and June 16, 2010, respectively. It contains 500 pairs of RSIs with a size of 50 × 50 pixels, which is based on the Dubai area for depicting urbanization changes. Each group of the two RSIs includes the "before" and "after" images and five different change descriptions. We split the dataset into training, validation, and testing sets following the method [24]. Specifically, 300 pairs of images are split for training, 50 pairs for validation, and 150 pairs for testing. Based on the original dataset, the RSI is further upsampled to a size of 256 × 256 before feeding them into a CNN-based feature extractor.

LEVIR-CC Dataset [20] is a publicly available RSICC dataset with significant land use changes (i.e., new buildings), providing a time span from 5 to 14 years. The RSI pairs are obtained from the Google Earth API, including 637 pairs with a high spatial resolution of 0.5 m/pixel. Each image has a fixed size of 1024 × 1024 pixels. There are different types of changed building instances, such as villa residences, tall apartments, small garages, and large warehouses. Each image is cropped into 16 nonoverlapping patches with a size of 256 × 256 pixels. Notably, each image pair is artificially annotated with 5 descriptive sentences for characterizing the changes. In this way, the LEVIR-CC dataset can provide 10 077 image pairs, containing changes or no changes. We follow the split [24]: 6815 pairs for training, 1333 pairs for validation, and 1929 pairs for testing.

2) Evaluation Metrics: To evaluate the performance of our proposed model and the ablated models, we utilize common evaluation metrics for RSIC task to report experimental results: BLEUn [42], ROUGE_L [43], METEOR [44], and CIDEr [45]. The derivative metric for BLEUn contains BLEU1, BLEU2, BLEU3, and BLEU4. These metrics are defined for n-gram precision, in which the ground-truth matchings are contained in the n-gram inferred results. Correspondingly, ROUGE_L seeks a recall ratio over the shared subsequence with maximum length matching the ground-truth term in the same order. METEOR extends the co-occurrence concept of BLEUn, considering the accuracy and recall of the entire corpus. Compared with BLEUn and ROUGE_L commonly used in machine translation, CIDEr is the only metric designed for RSIC task, it utilizes TF-IDF to assign different weights on different n-grams. The weights of frequently occurring phrases are lower, while infrequently occurring phrases are more special (with greater weights). The higher its value, the better. In this work, CIDEr will be attracted with high interest.

3) Implementation Details: In our experiments, the proposed network is implemented using the PyTorch on an Nvidia GeForce GTX 1080Ti GPU. The parameters of ResNet101 pretrained on ImageNet [37] and other blocks are optimized by the Adam optimizer [46] with initial learning rate of 1×10^{-4} and 5×10^{-4}, respectively. The initial learning rate of other blocks will be annealed by a factor of 0.8 for every 6 epochs. The training process will stop early by observing whether the BLEU4 score of the validation set increases within ten epochs. The maximum epoch is set to 35 epochs, and the batch size is set to 32 for the two public RSICC datasets. During the validation, the beam size is set to 3 for a better generation.

For each RSI pair, parallel ResNet101 is applied to detect original temporal features with dimension 2048. The word embedding size is fixed for 512. We set the maximum length of sentences in each paragraph to 50. The number of heads

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in MHA is 8 in all relevant modules. A single transformer decoder is adopted in our model. The vocabulary sizes for each dataset are 328 and 463 words for the Dubai-CC Dataset and LEVIR-CC Dataset, respectively.

4) Compare Baselines: In this section, we evaluate our model by comparing it with several state-of-the-arts approaches, which will be reviewed briefly as below.

1) *DUDA [13]:* DUDA first captures the representation of change by subtracting temporal CNN features. Dual spatial attention is integrated to select the discriminative change features, then translates to natural language using basic LSTM.
2) *MCCFormers [14]:* Two transformer change encoders (i.e., MCCFormers-D and MCCFormers-S) are designed for detecting multiple changes, transformer structure is used for predicting the changes words one by one.
3) *RSICCformer [20]:* RSICCformer designs a dual-branch transformer encoder to capture and recognize multiple changes of interest, a transformer decoder generates sentences for describing the differences.
4) *Chg2Cap [21]:* An attentive encoder combines hierarchical self-attention blocks with a residual block for attending change features, transformer-based sentence generator projects occurred change into logical sentence.
5) *SEN [25]:* SEN leverages a pretrained single-stream extractor for bi-temporal features and a designed difference feature encoder to enhance difference modeling explicitly during the change caption-generation process.
6) *ICT-Net [26]:* ICT-Net consists of a dual multiscale feature extractor and an interactive change-aware encoder, capturing the semantic changes between bi-temporal features. The transformer network with cross gated-attention is used to generate better change descriptions.

B. Evaluation Results and Analysis

Tables I and II show the performances of our model and detailed results from all competitors on the LEVIR-CC and Dubai-CC datasets. The compared methods focus on improving the difference feature encoder or language decoder while limiting symmetric difference feature modeling in dual-stream or single-stream extractors. Therefore, it is not hard to find that our proposed method achieves higher scores on all metrics compared with other methods on two datasets. This is because the bi-temporal features encoded by the CTA module model comprehensively understand the semantic relationships around objects in original temporal features. With the symmetric difference representations obtained by the SDT block, strengthening change representation ability can be easily constructed. Moreover, the bi-directional triplet ranking loss can effectively restrict the symmetric difference learning to improve the robustness. Thus, our model manifests its superior capability to handle the relevant change actively, which has boosted the performance of sentence generation to a certain extent.

1) Results on LEVIR-CC Dataset: As shown in Table I, our method performs on seven metrics with the previous RSICC methods DUDA [13], MCCFormers [14], RSICCformer [20], Chg2Cap [21], SEN [25], and ICT-Net [26]. Subtracting temporal CNN features for change features, DUDA gets really the lowest scores on all metrics. It is clear that the traditional change detection technology causes the absence of discriminative change feature representation, which reflects the unsatisfactory quality of change descriptions. Incorporating with single or double change transformer encoder, MCCFormers surpass DUDA across most metrics. However, it is also notable that MCCFormers-S performs favorably against the MCCFormers-D, indicating the effectiveness of the single change encoder on the LEVIR-CC dataset. Since the RSICCformer discovers adaptively multiple changes with stacked dual-branch transformer encoders, it overcomes some limitations that exist in MCCFormers-D. As we can see in Table I, RSICCformer improves CIDEr about 4.55% (131.40% vs 126.85%) compared with the MCCFormers-D, which is a significant margin. This means that a more efficient semantic change representation is needed for a more robust RSICC task. In our model, the CTA module aims to learn meaningful semantic relationships from changes involving distractions and generated coarse-to-fine change features by an SDT block. With sufficiently transformer-based attentive encoder, the masked change areas under Chg2Cap would maintain effect as much as possible, which gets really higher scores in Table I compared with RSICCformer. Worthy of note is that using pretrained single-stream extractor in SEN is also enough for the RSICC task, it is slightly inferior to Chg2Cap with the dual-stream extractor, which aims to mitigate the domain gap and high computational cost. Since the ICE-Net discovers multiscale discriminative change-aware information with a multilayer module, it surpasses all other methods across all metrics except for our model as champion. This is because changing features at each layer is a coarse-to-fine procedure. Furthermore, building upon symmetric change feature detection with temporal dependency, our network consistently outperforms ICE-Net with CIDEr 143.71% and ROUGE_L 76.64%. This is because our model is fed with more high-level semantic differences when generating words. Table I clearly reports that the symmetric difference shows better performance than directional change detection.

2) Results on Dubai-CC Dataset: The evaluation indicator results on the Dubai-CC dataset of the comparative experiments are shown in Table II. According to Table II, it can be found that MCCFormers-S is far less effective at identifying occurred changes than DUDA, in which a common subtraction is used for change features. This infers that subtraction can judge change features better than a single transformer encoder on RSI pairs in the Dubai-CC dataset. Among parallel transformer encoder models (i.e., MCCFormers-S and RSICCformer), the results of the RSICCformer are near to the MCCFormers-S. This is because the change representation extracted by the parallel encoder is more general, which can learn change features rather than locate change-related features. Chg2Cap tackles the problem with their attentive encoder to guarantee the strong representation ability for locating change regions, outperforming RSICCformer by about 25.95% on CIDEr and 6.70% on ROUGE_L. The scores of SEN in Table II are also a bit behind compared with

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TABLE I
PERFORMANCE COMPARISON OF OUR FRAMEWORK AND STATE-OF-THE-ART METHODS ON LEVIR-CC DATASET

Methods	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
DUDA	0.8144	0.7222	0.6424	0.5779	0.3715	0.7104	1.2432
MCCFormers-S	0.8216	0.7295	0.6542	0.5941	0.3826	0.7210	1.2834
MCCFormers-D	0.8049	0.7111	0.6352	0.5734	0.3823	0.7140	1.2685
RSICCformer	0.8411	0.7540	0.6801	0.6193	0.3879	0.7302	1.3140
Chg2Cap	0.8614	0.7808	0.7066	0.6439	0.4003	0.7512	1.3661
SEN	0.8510	0.7705	0.7001	0.6409	0.3959	0.7457	1.3602
ICT-Net	0.8606	0.7812	0.7145	0.6612	0.4051	0.7521	1.3836
Ours	**0.8679**	**0.7935**	**0.7310**	**0.6793**	**0.4160**	**0.7664**	**1.4371**

TABLE II
PERFORMANCE COMPARISON OF OUR FRAMEWORK AND STATE-OF-THE-ART METHODS ON DUBAI-CC DATASET

Methods	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
DUDA	0.5885	0.4359	0.3363	0.2539	0.2205	0.4834	0.6278
MCCFormers-S	0.5297	0.3702	0.2762	0.2257	0.1864	0.4329	0.5381
MCCFormers-D	0.6465	0.5045	0.3936	0.2948	0.2509	0.5127	0.6651
RSICCformer	0.6792	0.5361	0.4137	0.3128	0.2541	0.5196	0.6654
Chg2Cap	**0.7204**	0.6018	0.5084	**0.4170**	0.2892	0.5866	0.9249
SEN	0.7095	0.5728	0.4581	0.3625	0.2662	0.5595	0.9177
ICT-Net	0.6938	0.5703	0.4650	0.3617	0.2678	0.5731	0.9297
Ours	0.7181	**0.6060**	**0.5088**	0.4137	**0.2967**	**0.6324**	**1.0947**

Chg2Cap. While the excellent performance of the ICT-Net on the Dubai-CC dataset infers that the multilayer interactive change-aware module facilitates the discernment of critical changes for a better RSICC network. Focusing on robust symmetric change detection, we can observe that our model achieves superior performance both on CIDEr and ROUGE_L, with 109.47% and 63.24%, which confirms its effectiveness. It shows the importance role of the SDT block, which accepts temporal information obtained by our CTA module and bears the responsibility for the temporal difference representation. Therefore, the advantages of the symmetric difference can be effectively integrated into the RSICC task.

We also compared the parameters and inference speed (images per second) of our model with more recent methods on the Dubai-CC dataset. The comparison results are shown in Table III. The Chg2Cap has the fewest parameters, thus the testing process can understand more images per second. Although the a higher number of parameters in the designed SDT, our inference speed is slightly higher than ICT-Net and RSICCformer.

We show visualization examples of the attention maps from our model in Fig. 5(a) and (b) together with their generated change captioning results. The LEVIR-CC dataset has super high resolution and can clearly identify changed objects such as buildings on the ground. It can be seen that our framework successfully responds to the ground truths and focuses on the change in visual contents described in the corresponding sentence. Interestingly, the correlations between the attended regions in the "after" RSI and predicted key words are obvious. This verifies the advantage of our model that revolves around the acquisition and aggregation of symmetric differences learning in the full model.

TABLE III
COMPARISON OF OUR FRAMEWORK AND STATE-OF-THE-ART METHODS IN TERMS OF TRAINING TIME AND INFERENCE SPEED (IMAGES PER SECOND). ALL RESULTS ARE REPORTED BASED ON DUBAI-CC DATASET

Methods	Parameters (M)	Inference speed
RSICCformer	56.20	12.67
Chg2Cap	32.81	39.58
SEN	39.90	3.79
ICT-Net	96.40	10.44
Ours	91.40	11.61

C. Ablation Experiments

To verify the innovations of our network, we conduct ablation studies to validate the effectiveness and contributions of proposed components. The Baseline with Siamese ResNet101 extracts original temporal features, following direct subtraction for change features. Then, the one-layer transformer is mainly responsible for sentence generation. The ablative results are established on the LEVIR-CC and Dubai-CC datasets. As can be seen from Tables IV and V, Baseline is a sufficiently strong baseline model, which also illustrates our approach more reliable.

1) Effectiveness of CTA: To make better use of the characteristics of bilateral features, the CTA module in our network derives intertemporal information interaction and performs task-relevant feature aware before change feature detection, which improves the performance on the bias of the Baseline model. As presented in Table IV, the utilization of CTA produced an increase of greater than 3.82% (133.98% versus 130.16%) in terms of CIDEr. Especially, 10.82% (87.27%

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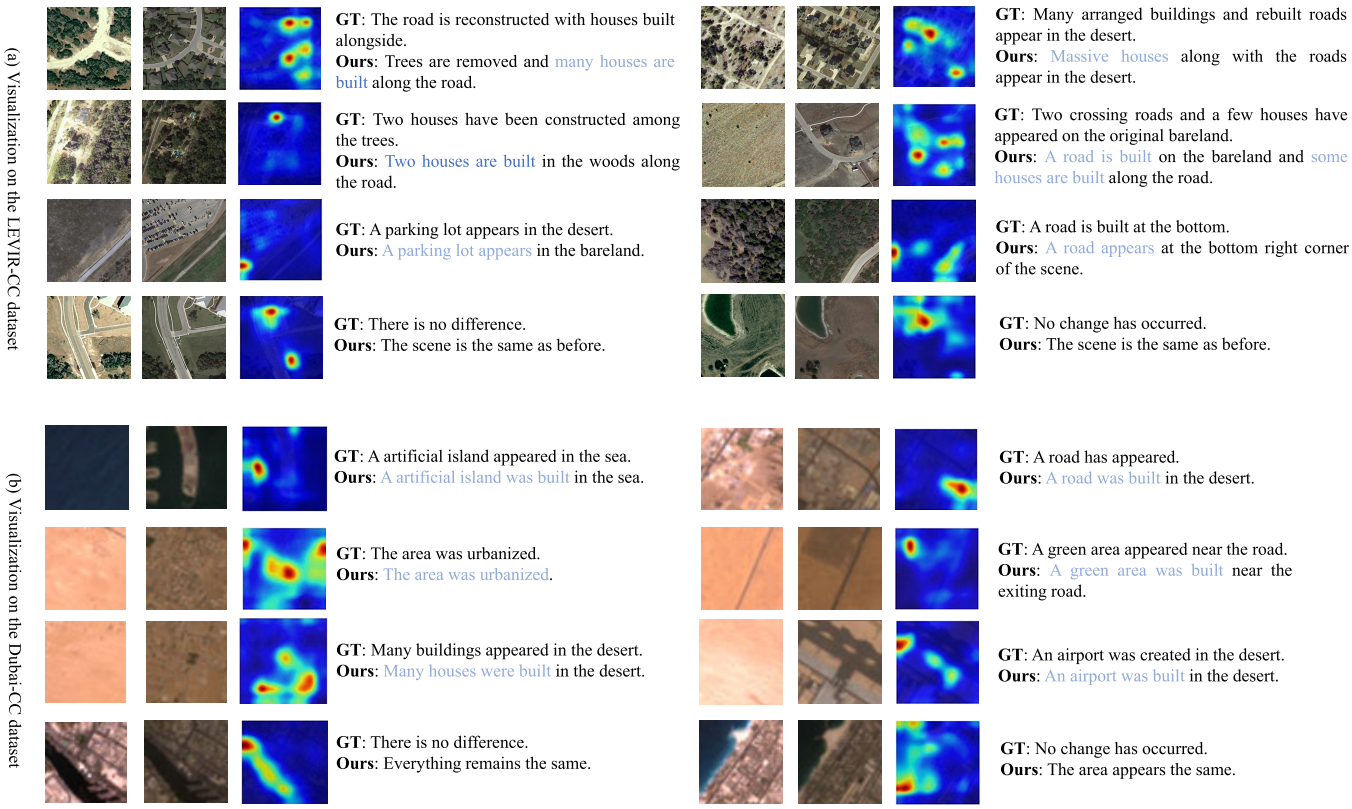


Fig. 5. Visualized attention maps in "after" RSI and difference description examples generated by our model on (a) LEVIR-CC dataset and (b) Dubai-CC dataset. The blue words indicate changes in the input RSI pair.

TABLE IV
ABLATION PERFORMANCE OF OUR SIN ON LEVIR-CC DATASET

Methods	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
Baseline	0.8329	0.7482	0.6800	0.6254	0.3858	0.7198	1.3016
CTA	0.8412	0.7643	0.7008	0.6503	0.3972	0.7353	1.3398
CTA+SDT	0.8647	0.7884	0.7240	0.6715	0.4076	0.7568	1.3951
Full	**0.8679**	**0.7935**	**0.7310**	**0.6793**	**0.4160**	**0.7664**	**1.4371**

TABLE V
ABLATION PERFORMANCE OF OUR SIN ON DUBAI-CC DATASET

Methods	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
Baseline	0.6982	0.5626	0.4438	0.3425	0.2652	0.5597	0.7645
CTA	0.7000	0.5677	0.4501	0.3337	0.2814	0.6089	0.8727
CTA+SDT	0.7147	0.5898	0.4808	0.3817	0.2946	0.6188	1.0051
Full	**0.7181**	**0.6060**	**0.5088**	**0.4137**	**0.2967**	**0.6324**	**1.0947**

versus 76.45%) higher CIDEr score than Baseline is shown in Table V. It indicates the change-related information contained in original temporal features is capable of providing strong task-relevant features for benefiting difference understanding.

2) Effectiveness of SDT: With the aim of isolating the performance gained from symmetric change features, the SDT block is embedded in the first ablative model. There is temporal symmetry to enforce networks for learning nondirectional difference representations, which is an improved version of change detection by simple subtraction. As can be seen in Table IV, 5.53% higher CIDEr and 2.15% higher ROUGE_L scores than the ablative CTA model. Table V shows the performance of the ablation model on the Dubai-CC dataset. The results are similar to those in Table IV, so we can get a similar conclusion: 13.24% higher CIDEr and 0.99% higher ROUGE_L scores than the ablative CTA model. Thus, the SDT block bridges interaction based on multilevel temporal features and leads to obvious improvements.

3) Effectiveness of Ranking Loss: The full model performs the symmetric supervision of nondirectional change representation with bi-directional triplet ranking loss. The ablative experiments in Tables IV and V indicate that our model is inclined to propagate offsets against the current symmetric change features to adjust dynamic difference learning process.

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TABLE VI
EFFECT OF DIFFERENT SIAMESE BACKBONES ON LEVIR-CC DATASET

Backbones	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
ResNet18	0.8231	0.7333	0.6503	0.5711	0.3639	0.7403	1.3645
ResNet34	0.8407	0.7645	0.6999	0.6462	0.3928	0.7471	1.3959
ResNet50	0.8590	0.7808	0.7148	0.6589	0.4035	0.7593	1.4258
ResNet101	**0.8679**	**0.7935**	**0.7310**	**0.6793**	**0.4160**	**0.7664**	**1.4371**

TABLE VII
EFFECT OF DIFFERENT SIAMESE BACKBONES ON DUBAI-CC DATASET

Backbones	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
ResNet18	0.6935	0.5592	0.4414	0.3309	0.2620	0.5504	0.7936
ResNet34	0.6930	0.5869	0.4935	0.4035	0.2840	0.5857	0.9273
ResNet50	**0.7367**	0.6013	0.4904	0.3819	0.2913	0.6217	0.9946
ResNet101	0.7181	**0.6060**	**0.5088**	**0.4137**	**0.2967**	**0.6324**	**1.0947**

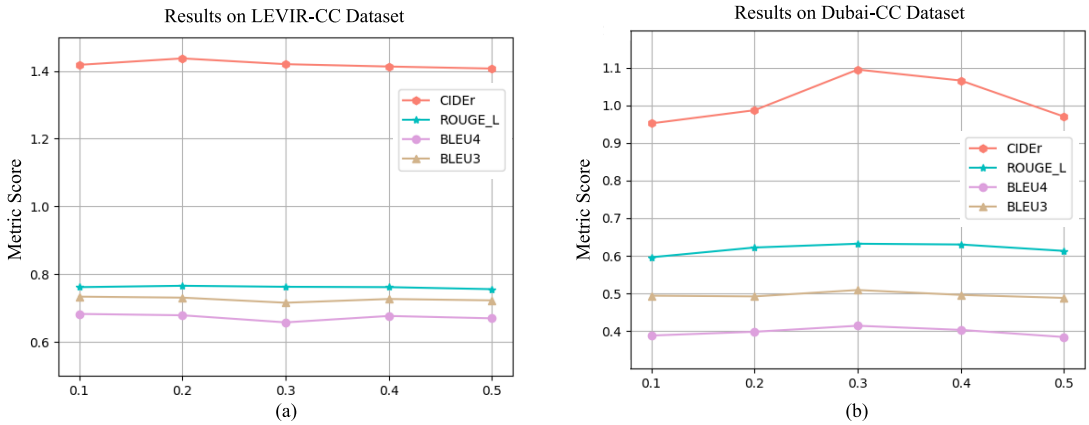


Fig. 6. Performance visualizations of our model (i.e., BLEU3, BLEU4, ROUGE_L, CIDEr) that are affected by parameters with different parameters γ on LEVIR-CC and Dubai-CC datasets. (a) Setting different γ. (b) Setting different γ.

To be specific, the full model with bi-directional triplet ranking loss produces better results than the "CTA + SDT" model. For the LEVIR-CC test set, the SIN can steadily improve the CIDEr at most 4.20% and 0.96% based on ROUGE_L. For the Dubai-CC test set, it can also boost the CIDEr with consistent gains of about 8.96% and 1.36% upon ROUGE_L.

The good performance of the proposed components can be seen from the aforementioned experiments and comparisons. Besides, it is observable that the performance improvements on the Dubai-CC dataset are higher than the LEVIR-CC dataset. The manually annotated sentences in the LEVIR-CC dataset will be more semantically-rich, which may bring more difficulties for training the models. This phenomenon demonstrates that the results are sensitive to the size of the dataset.

D. Parameter Analysis

In recent years, different ResNets have been widely used in computer vision, each with its own advantages and focus. In our model, Siamese ResNet101 is responsible for extracting original temporal features. To verify the effectiveness of different Siamese ResNets, we investigate the performance of our method with respect to different parallel backbones. By cascading the bi-directional triplet ranking loss and the proposed method, the framework can be trained as a whole. There is a variable offset parameter γ added to L_r. We will make a sensitive analysis for important parameters. All experiments are conducted on the LEVIR-CC and Dubai-CC datasets.

1) Influence of Siamese ResNet Architectures: To ensure the fairness of the comparison, we use different Siamese ResNets with other settings same in all experiments. The performances on LEVIR-CC and Dubai-CC datasets are shown in Tables VI and VII separately, combining different original temporal features learned from Siamese ResNet18, ResNet34, ResNet50, and ResNet101. On the LEVIR-CC dataset, our model based on ResNet101 obtains the highest scores. As shown in Table VI, a better performance could be obtained by stacking more deeper ResNet structures. It is worth noting that when Siamese ResNet34 is adopted, our model is also competitive compared with the Chg2Cap model. Besides, we also compare the results of our model on the Dubai-CC dataset. As can be seen from the comparison in Table VII, the depth of ResNet is greatly increased, it also brings a progressive improvement in evaluating metrics for sentence generation. Our model gets the highest scores benefiting from ResNet101 in terms of overall metrics, which indicates the competitiveness of the proposed structure. Obviously, injecting an excellent ResNet backbone with our model demonstrates much better performance than

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the unfiltered one. The differences with different Siamese ResNets further reflect the flexibility of our method, which also enables our network to remain optimal in most metrics.

2) Sensitivity of Hyperparameter γ: To ensure the fairness of the comparison, we use Siamese ResNet101 with different γ settings. The offset parameter γ added to L_r in our model is an important hyper-parameter, in which L_r builds symmetric supervision between un-directional change features. So, it can provide effective assistance for such a more reasonable structural design. We compare several settings on the LEVIR-CC and Dubai-CC datasets in Fig. 6. As shown in Fig. 6(a), compared with other settings, "$\gamma = 0.2$" has the best performance on CIDEr metric, so it is used as the default setting in all experiments on LEVIR-CC dataset. For experiments on the Dubai-CC dataset shown in Fig. 6(b), the "$\gamma = 0.3$" is as default setting, which leads to a better performance on seven metrics. An interesting phenomenon occurs under the γ settings on the LEVIR-CC and Dubai-CC datasets. Taking the LEVIR-CC dataset as an example, other performances in Fig. 6(a) are not as good as the setting "$\gamma = 0.2$," but it is still competitive than Chg2Cap model. This also reflects that simply increasing the γ setting does not necessarily lead to a better performance, which plays an important role in competitive performance. Thus, the value of γ should be tuned carefully.

V. Conclusion

In this article, we present a novel approach to generate a natural-language description of what has changed in a pair of RSIs. At first, the RSI pair is encoded by a feature extractor, providing "before" features and "after" features. We consider that the cross-temporal feature relation learning is more effective than two parallel individual feature learning for the RSICC task. Motivated by this, a CTA algorithm is designed to perceive changes in cues and suppress irrelevant variations in intramodal features. Generally, accessing difference features relies on simple subtraction operations on obtained temporal features, which are not reliable change maps. The SDT block extracts symmetric change maps exploring semantic change representation. Furthermore, bi-directional triplet ranking loss is encouraged for matching symmetric change maps, which is essentially to match changed pixels and unchanged pixels. Overall, numerical results validate that our method can achieve competitive or even superior performance compared with state-of-the-art methods on two public datasets. In addition, semantic change detection applied in the RSICC task is efficient for change prediction, this evidence deserves further study in the future.

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